Shreyas Kumar

AI Engineer | LLM Engineer | Applied ML Engineer

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EDUCATION

SYRACUSE UNIVERSITY | MS in Computer Science | GPA: 3.6/4 — Aug 2023 – May 2025

KIIT | B. Tech in Computer Science | GPA: 9.1/10 — Jun 2019 - Apr 2023

WORK EXPERIENCE

Syracuse University | Research Assistant | Syracuse, NY — Jun 2024 – Jun 2026

- Processed 500K+ annotated ORACC tokens into optimized **Hugging Face datasets** through **parallelized preprocessing pipelines**, reducing data preparation time ~40% for downstream **NLP** experimentation.
- Combined **BM25 lexical retrieval** with **768-d dense embeddings (LangChain)** to improve query relevance ~30% vs baseline keyword search, **validated using Recall@K** and **Mean Reciprocal Rank (MRR)**.
- Applied **LoRA fine-tuning** to **Microsoft Phi**, reducing VRAM consumption ~50% while maintaining sub-second CPU inference and stable response quality.
- Exposed the **RAG** pipeline as a reusable **FastAPI + Docker microservice** with CI via **GitHub Actions**, continuous **p50/p95 latency monitoring**, and **automated retrieval regression tracking**.

NielsenIQ | Software Automation Engineer Intern | Pune, IN — Oct 2022 – Jun 2023

- Automated UI validation using **Selenium WebDriver + JavaScript**; authored and executed **100+ regression/integration test cases**, reducing daily runtime from **7–8 hours to 3 hours**.
- Monitored daily test runs, analyzed reports, partnered with developers to resolve defects within 24 hours, and integrated suites into CI with improved reporting/triage to shorten regression turnaround and raise release confidence.

PROJECTS

Lumo: Child-safe Voice AI Listening Companion [Link] — Jan 2025 – Present

- Architected **low-latency (<300ms) parent-controlled voice companion for children** using OpenAI realtime APIs (semantic VAD, streaming STT/TTS) over FastAPI WebSockets with PCM16 streaming & event-driven session orchestration.
- Designed **deterministic single-writer kernel** (event queue + state machine) enforcing strict turn-taking and at-most-one assistant response per turn, reducing race conditions and duplicate emissions by ~90%.
- Built a **privacy-first safety system** with transcript-only classification, explainable decision codes, structured parent escalation, and **no raw-audio storage**, ensuring full compliance with child data minimization principles.
- Containerized backend with Docker Compose and shipped a Vite client supporting autoplay, **<150ms interruption (barge-in) response time**, correlation IDs for turn tracking, and end-to-end latency instrumentation across sessions.

ReportXplain: HenHacks 25' — Mar 2025

- Implemented a **Gemini**-powered extraction pipeline to summarize structured and unstructured medical reports (~10s/report), surfacing key findings and improving patient accessibility to clinical information.
- Architected secure **NestJS + PostgreSQL APIs** for ingestion and processing, integrated **Docker/MinIO** object storage, and PDF parsing with interactive upload visualizations.

AgroVision - Smart Farming Solution: HackMerced 25' — Mar 2025

- Designed a real-time agricultural analytics platform combining drone imagery, NPK/humidity sensor streams, and **OpenWeather** feeds **(Next.js + FastAPI/Celery + MongoDB + Docker)** for soil moisture and plant-health monitoring.
- Integrated **Letta + Gemini** to generate crop recommendations, carbon-footprint insights, and livestock-tracking alerts, reducing manual scouting 50% and securing **2nd place in Ag-Tech**.

Cuse-Rank | FIRST PRIZE WINNER — Feb 2025

- Developed a judge allocation engine with **OR-Tools constraint optimization** to improve fairness, workload balance, and scheduling efficiency for research evaluations.
- Orchestrated an **async FastAPI + Celery pipeline** with **PostgreSQL queues**, cutting runtime 50% via parallel tasks and query tuning.
- Implemented **embedding-based similarity scorer** for judge–poster matching using professor expertise vectors and abstracts.

SKILLS

LLM / NLP: RAG, LangChain, Hugging Face, Transformers, LoRA, BM25, Embeddings, Prompt Engineering, STT/TTS
ML / Data: PyTorch, JAX, XGBoost, Random Forest, MLflow, Model Evaluation
Backend / APIs: FastAPI, Celery, PostgreSQL, SQLAlchemy, Redis, Node.js, NestJS, WebSockets, Asyncio
Deployment / Tools: Docker, Kubernetes, GitHub Actions, Azure ML, Heroku, Firebase, Selenium, MinIO
Languages: Python, Rust, Java, Kotlin, TypeScript/JavaScript, SQL, SwiftUI